

1.2.3 Units

☐ The units of data storage:

- Bit
- Nibble (4 bits)
- Byte (8 bits)
- Kilobyte (1,000 bytes or 1 KB)
- Megabyte (1,000 KB)
- Gigabyte (1,000 MB)
- Terabyte (1,000 GB)
- Petabyte (1,000 TB)

☐ How data needs to be converted into a binary format to be processed by a computer

☐ Data capacity and calculation of data capacity requirements

Required

- ✓ Why data must be stored in binary format
- ✓ Familiarity with data units and moving between each
- ✓ Data storage devices have different fixed capacities
- ✓ Calculate required storage capacity for a given set of files
- ✓ Calculate file sizes of sound, images and text files
 - $\text{sound file size} = \text{sample rate} \times \text{duration (s)} \times \text{bit depth}$
 - $\text{image file size} = \text{colour depth} \times \text{image height (px)} \times \text{image width (px)}$
 - $\text{text file size} = \text{bits per character} \times \text{number of characters}$

Alternatives

- Use of 1,024 for conversions and calculations would be acceptable
- Allowance for metadata in calculations may be used